COMFORT-CENTRIC TRAJECTORY PREDICTION FOR AUTONOMOUS DRIVING VIA TEMPORAL GRAPH CROSS ATTENTION

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ABSTRACT

Advanced autonomous driving (AD) systems are expected to go beyond collision avoidance by considering passenger ride comfort, traffic flow stability, and user trust. Reactive control often leads to frequent and abrupt interventions, which can degrade comfort and undermine confidence in autonomous operation, highlighting the need for predictive driving strategies that anticipate surrounding vehicle behavior. While prior studies have mainly focused on surrounding or rear vehicles, proactive anticipation of front vehicle behavior remains critical in interactive traffic environments. This study proposes a front vehicle trajectory prediction framework that models spatiotemporal interactions as graph representations extracted from monocular dashcam video and forecasts future trajectories using a Temporal Graph Cross Attention (TGCA) mechanism. TGCA selectively emphasizes behaviorally influential vehicles by capturing temporally aligned interaction patterns across consecutive frames. We validated the feasibility and applicability of the proposed framework through real-world dashcam experiments. The proposed model achieved a 5-second prediction MAE of approximately 0.09. These results indicate that front vehicle trajectory prediction can serve as a practical and effective foundation for prediction-driven anticipatory driving strategies in future AD systems operating under realistic sensing constraints.

INTRODUCTION

Advanced autonomous driving (AD) systems must evolve beyond basic collision avoidance to provide a comprehensive driving experience that ensures passenger ride comfort, traffic flow stability, and user trust (1, 2). While current technologies have reached technical maturity (3, 4), they primarily rely on reactive control. In real-world environments, this often results in abrupt interventions—such as sudden deceleration during cut-in scenarios—which degrade comfort, induce anxiety, and disrupt surrounding traffic flow. To realize stable and natural driving behavior (5), AD systems require an anticipatory capability to recognize interaction

patterns early and adjust strategies before hazards fully materialize.

Motivated by this need, this study proposes a proactive anticipatory driving framework that predicts the future trajectories and maneuvers of front vehicles. By explicitly modeling spatio-temporal interactions using monocular dashcam video, the system enables the ego vehicle to determine gradual, smooth responses in advance, thereby reducing sudden control interventions and suppressing the propagation of traffic disturbances.

The proposed framework consists of three main components. First, vehicle states extracted from dashcam videos are structured into spatio-temporal graphs, where nodes represent detected vehicles and edges encode their spatial relationships relative to the ego vehicle. Second, the Temporal Graph Cross Attention (TGCA) mechanism serves as the core component to capture time-dependent interaction dynamics, selectively emphasizing behaviorally influential vehicles by jointly considering their temporal evolution and spatial relevance. Finally, the predicted trajectories are interpreted at the maneuver level to directly support the ego vehicle's anticipatory decision-making.

The feasibility and effectiveness of the TGCA-based framework were validated through real-world dashcam experiments. The results demonstrate the model's superior trajectory prediction accuracy and robustness in capturing dynamic spatial interactions. By operating under realistic sensing conditions without relying on high-cost infrastructure, this prediction-driven approach establishes a practical foundation for real-world deployment in future AD systems.

RELATED WORK

ANTICIPATORY DRIVING AND COMFORT-CENTRIC SAFETY

Traditionally, safety in advanced autonomous driving (AD) systems focused primarily on functional collision avoidance, but contemporary environments increasingly require user-centric considerations where safety and passenger ride comfort are intrinsically linked. Because reactive maneuvers can significantly heighten passenger anxiety and erode trust, recent research has prioritized early risk prediction and the suppression of abrupt interventions to realize smoother, more predictable driving behaviors (6, 7). Collectively, these findings establish that gradual interventions initiated at the stage of emerging risk cues are far more effective than immediate reactions to explicit hazards.

Concurrently, various decision-making and trajectory planning approaches have been proposed to jointly address safety and comfort (8, 9). However, these existing works predominantly rely on structured numerical trajectory data, external priors (such as connectivity or maps), or high-cost sensor suites like LiDAR and HD maps. Consequently, they are inherently limited in their ability to capture fine-grained behavioral cues from real-world driving videos, constraining both scalability and cost-effectiveness for mass-market ADAS deployment.

In contrast, our work positions itself as a vision-only, interaction-aware prediction framework. By specifically focusing on the lead and adjacent vehicles most relevant to near-term ego stability, the proposed framework exploits early visual cues to enable earlier and smoother responses under realistic sensing constraints.

VEHICLE TRAJECTORY PREDICTION

Vehicle trajectory prediction serves as a cornerstone for decision-making and motion pl anning in autonomous driving systems, necessitating the precise modeling of spatio-tem poral interactions and human uncertainty. To achieve this, recent advancements have e xtensively leveraged Graph Neural Networks (GNNs) and attention mechanisms to expl icitly model inter-agent interactions and structure spatio-temporal dependencies (10, 11). Concurrently, persistent efforts have integrated high-level information—such as driver intent, structural road topology, and obstacles—to generate context-appropriate trajectori es (12), while generative models have been proposed to address multi-modal uncertaint y through probabilistic inference (13).

Despite these significant advancements, existing models typically treat all surrounding objects uniformly or rely on global scene representations. Consequently, they often fail to sufficiently reflect the differential importance of leading or adjacent vehicles, which exert the most direct influence on the ego vehicle's immediate driving stability and ride comfort. Furthermore, most state-of-the-art pipelines are predicated on structured numerical trajectory inputs, high-cost LiDAR-centric perception, or HD-map priors. This reliance imposes structural limitations on directly utilizing fine-grained early visual cues from real-world videos and fundamentally constrains scalability in cost-sensitive ADAS scenarios.

To address these limitations, this study proposes a TGCA-based trajectory prediction framework utilizing solely monocular dashcam video to model interactions between the ego vehicle and leading/lateral vehicles as a spatio-temporal graph. By cross-referencing changes in graph states over time, the architecture precisely captures subtle behavioral shifts and selectively emphasizes key agents that substantially impact short-term driving decisions. Ultimately, this selective, mapless, and sensor-light modeling positions TGCA as a highly practical, vision-only alternative for interaction-critical, near-term forecasting under realistic sensing constraints.

PRELIMINARY

PROBLEM STATEMENT

This study focuses on predicting the trajectories of front vehicles—defined as those detected within the field of view of a forward-facing dashcam—as a foundational task to support anticipatory maneuver planning with a focus on passenger comfort. Because these vehicles significantly affect the ego vehicle's motion through sudden maneuvers, explicitly modeling their spatio-temporal interactions is essential.

To capture these interactions, we construct an undirected spatial graph, $\mathcal{G}$, at each time step. The node set, $\mathbf{X}$, represents the positions of the ego and front vehicles, while the edge set, $\mathbf{E}$, encodes the spatial interactions connecting them. By sequentially constructing this graph across consecutive frames, the resulting temporal graph sequence serves as the input to estimate future trajectories. Furthermore, unlike conventional studies that assume a stationary viewpoint, this framework utilizes a dynamic observation perspective from the moving ego vehicle, specifically incorporating relative distance and motion patterns into the prediction process. Finally, these predicted trajectories are used to support downstream anticipatory decision-

making, which is qualitatively examined through a simplified traffic simulation to assess the potential impact on overall traffic flow.

PROBLEM FORMULATION

We formulate the trajectory prediction problem for front vehicles by modeling spatio-temporal interactions as a temporal sequence of undirected graphs, defined as $\mathcal{G} = \{\mathcal{G}_t\}_{t=1}^{T}$. At each observation frame t, the spatial graph $\mathcal{G}_t = (\mathbf{X}_t, \mathbf{E}_t)$ captures the spatial states of the ego vehicle and $N-1$ front vehicles via the node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N_t \times d}$, while the edge set $\mathbf{E}_t$ explicitly models their spatial interactions. The objective is to forecast the future two-dimensional relative positions of these front vehicles over a prediction horizon from $T+1$ to P. To achieve this, the proposed model f_θ maps the observed graph sequence to the predicted trajectories, denoted as $\widehat{\mathbf{Y}} = f_\theta(\mathcal{G})$. Finally, the model is trained by minimizing a loss function $\mathcal{L}$ that measures the discrepancy between predicted and ground truth trajectories to obtain the optimal parameter set $\theta^* = \mathrm{argmin}_\theta \mathcal{L}(\theta)$. This structured formulation establishes the essential foundation for anticipatory and comfort-aware maneuver planning in autonomous driving systems.

PROPOSED METHOD

OVERALL ARCHITECTURE OF THE PROPOSED FRAMEWORK

The proposed framework consists of three primary components, as illustrated in Figure 1. Figure 1: (1) a video input and graph representation module, which extracts vehicle positional information and constructs a spatio-temporal graph structure; (2) a front vehicle trajectory

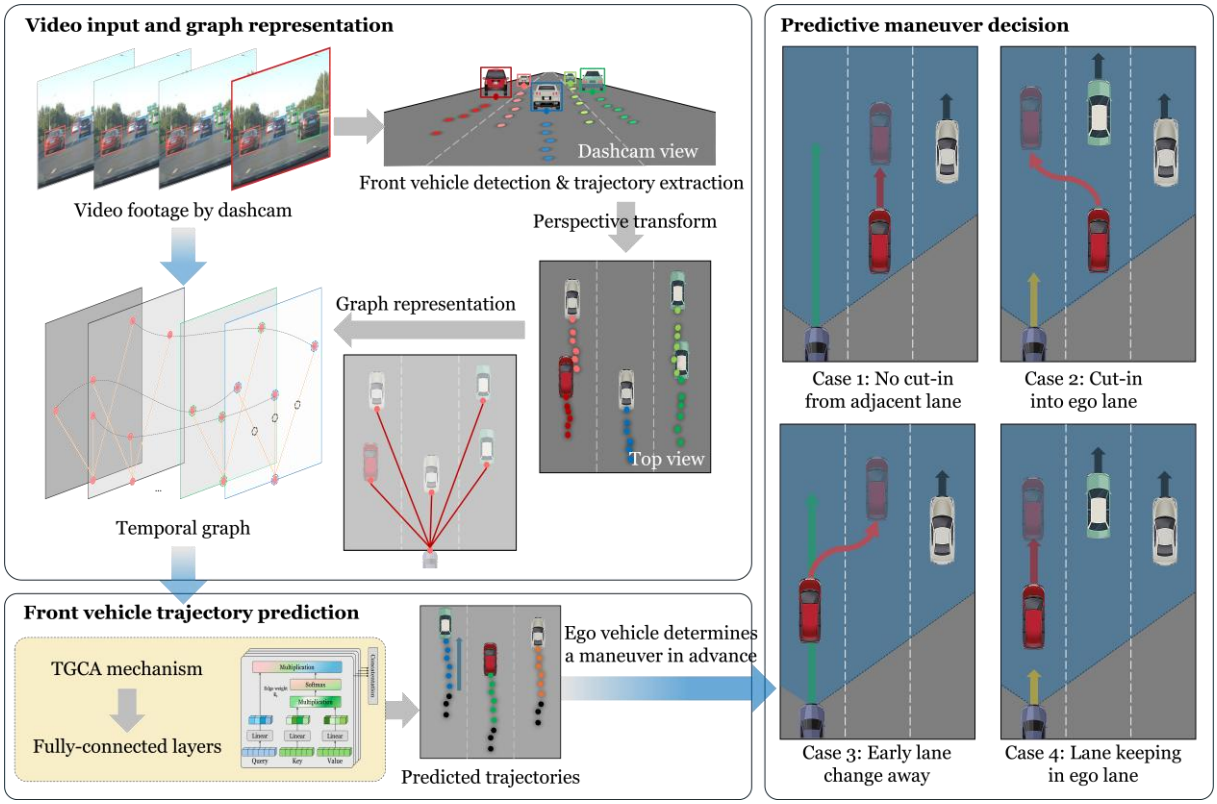


Figure 1. Overview of the proposed framework

prediction module that models time-dependent interactions using a TGCA mechanism; and (3) a predictive maneuver decision module that selects an ego vehicle response from the predicted front vehicle trajectories. The policy is rule-based and introduces no additional learnable parameters.

VIDEO INPUT AND GRAPH REPRESENTATION

The first stage of the proposed framework transforms raw forward-facing video into structured spatio-temporal graphs representing interactions between the ego vehicle and surrounding vehicles. This is accomplished via a computer vision pipeline that combines YOLOv11 (14) for object detection and ByteTrack (15) for tracking, followed by a perspective transformation into top-view coordinates for geometrically interpretable motion analysis (16, 17).

At each time step t, a spatial graph $\mathcal{G}_t = (\mathbf{X}_t, \mathbf{E}_t)$ is constructed. The node feature matrix $\mathbf{X}_t = \left[\boldsymbol{x}_t^0,\ \boldsymbol{x}_t^1,\ \ldots,\ \boldsymbol{x}_t^{N_t-1}\right]^\top$, $\boldsymbol{x}_t^i = \left(x_t^i, y_t^i\right) \in \mathbb{R}^2$ captures the top-view positions (x, y) of the ego vehicle (indexed as node 0) and the $N_t - 1$ front vehicles. Concurrently, the edge set $\mathbf{E}_t = \left\{\left(0, i, e_t^{0,i}\right) \mid i = 1, \ldots, N_t - 1\right\}, e_t^{0,i} = \left\|\boldsymbol{x}_t^0 - \boldsymbol{x}_t^i\right\|_2^2$ encodes spatial interactions utilizing the squared Euclidean distance between the ego vehicle and each front vehicle. To reflect spatial proximity and ensure numerical stability, these distances are normalized, incremented by a small constant ϵ, and inverted to obtain the interaction coefficient $\tilde{e}_t^{0,i}$. This transformation yields a rescaled vector $\tilde{\mathbf{e}}_t = [\tilde{e}_t^{0,0}, \tilde{e}_t^{0,1}, \cdots, \tilde{e}_t^{0,(N_t-1)}]^\top \in \mathbb{R}^{N_t}, \tilde{e}_t^{0,0} = 0$ that assigns larger weights to closer vehicles, excludes self-connections, and serves directly as a multiplicative modulator in the attention computation. Finally, all trajectories within a fixed Region-of-Interest (RoI) are represented in the top-view plane and processed using per-axis z-score standardization for all subsequent modeling and evaluation.

FRONT VEHICLE TRAJECTORY PREDICTION

Given the temporal graph sequence $\mathcal{G} = \{\mathcal{G}_t\}_{t=1}^T$, the primary objective of this module is to learn a predictive function f_θ capable of estimating the future positions of front vehicles over a prediction horizon from $t = T + 1$ to P. Central to this predictive modeling is the Temporal Graph Cross Attention (TGCA) mechanism. TGCA is designed to capture temporal interactions through a temporally directed cross-attention structure that operates in a sliding window fashion, generating predictions frame-by-frame by referencing vehicle states from the immediately preceding time step (see Figure 2).

At each prediction step t, the node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N_t \times 2}$, which contains the 2D position features (x, y) for N_t vehicles, is linearly projected into the attention space. Specifically, the queries are derived from the current time step, while the keys and values are extracted from the previous time step:

$$\mathbf{Q} = \mathbf{X}_t \mathbf{W}^Q, \quad \mathbf{K} = \mathbf{X}_{t-1} \mathbf{W}^K, \quad \mathbf{V} = \mathbf{X}_{t-1} \boldsymbol{W}^V \tag{1}$$

where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{2 \times d_h}$ represent the shared learnable projection matrices, and d_h denotes the hidden dimension.The core innovation of the TGCA mechanism lies in its integration of spatial interaction strengths directly into the attention computation. After calculating the scaled dot-product attention between the queries and keys, the resulting attention scores are explicitly modulated using the spatial edge weights $\tilde{\boldsymbol{e}}_t$ derived from the spatial graph $\mathcal{G}_t$. This modulation is performed via row-wise element-wise multiplication:

$$\mathbf{Z}_t = \text{softmax}\left(\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}\right) \odot (\tilde{\boldsymbol{e}}_t \mathbf{1}^\top)\right)\mathbf{V} \tag{2}$$

where $\odot$ denotes element-wise multiplication, and $\tilde{\mathbf{e}}_t \mathbf{1}^\top \in \mathbb{R}^{N_t \times N_{t-1}}$ is formed by repeating the edge weights across columns. Consequently, this formulation ensures that vehicles with larger proximity-based weights receive higher attention, enabling the model to selectively focus on the most behaviorally relevant agents during prediction.

To capture diverse interaction patterns, TGCA employs a multi-head attention framework. Each head computes its respective embeddings by cross-attending between consecutive frames t and $t-1$. The outputs from all H attention heads are then concatenated to form a comprehensive vehicle embedding:

$$\mathbf{Z}_t^{\text{final}} = \text{Concat}(\mathbf{Z}_t^1, \cdots, \mathbf{Z}_t^H) \in \mathbb{R}^{N_t \times (H \times d_h)} \tag{3}$$

Subsequently, each consolidated vehicle embedding is processed through fully-connected layers utilizing an ELU activation function. The final layer outputs a 2D vector representing the predicted future coordinates for each vehicle:

$$\hat{\mathbf{y}}_t^i = \text{FC}(\mathbf{Z}_t^{\text{final}}) \in \mathbb{R}^2 \tag{4}$$

This trajectory prediction process is conducted autoregressively to generate the complete sequence of future trajectories, denoted as:

$$\hat{\mathbf{Y}} = f_\theta(\mathcal{G}) = \{\hat{\mathbf{y}}_t^i\}_{t=T+1}^{P} \tag{5}$$

Finally, the learnable parameters θ of the model are optimized by minimizing the Mean Squared Error (MSE) loss, which precisely measures the discrepancy between the predicted and ground-truth trajectories.

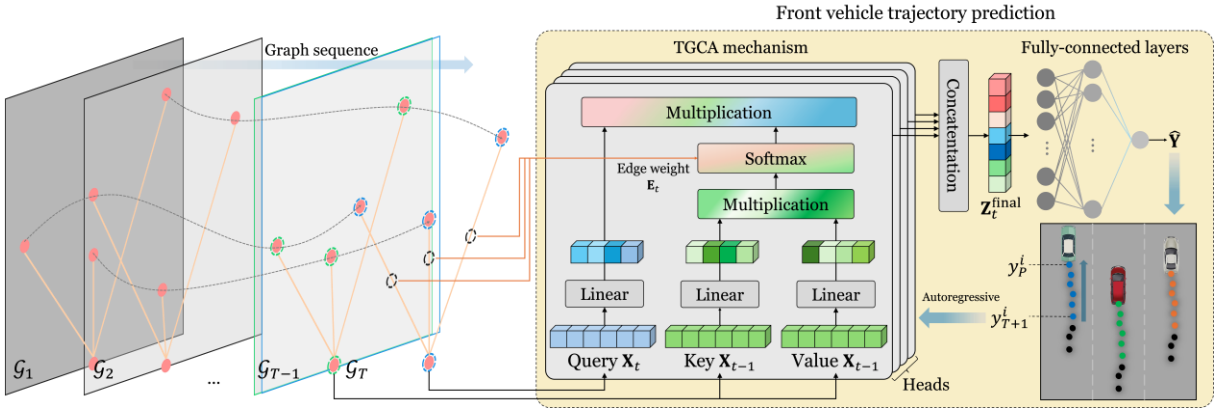


Figure 2. The proposed Temporal Graph Cross Attention (TGCA) model

PREDICTIVE MANEUVER DECISION

Upon obtaining the predicted future trajectories $\hat{\mathbf{Y}} = \{\hat{\mathbf{y}}_t^i\}_{t=T+1}^{P}$ for all front vehicles, the predictive maneuver decision module selects an ego vehicle response policy focused on ride comfort and traffic-flow stability. Rather than performing maneuver classification for surrounding vehicles, this stage translates trajectory forecasts directly into ego vehicle responses. To maintain interpretability and isolate the impact of trajectory prediction, the

module employs a simple rule-based decision policy without introducing additional learnable parameters.

The decision process evaluates the longitudinal and lateral displacement patterns of each front vehicle, alongside their relative positions and lane affiliations. Using trajectory-based heuristics such as accumulated lateral displacement, directional deviation, and projected lane alignment, the module infers four representative interaction scenarios to guide the ego vehicle's response:

- Case 1: No Cut-in. If an adjacent vehicle maintains its trajectory without lateral movement toward the ego lane, the ego vehicle maintains its current behavior to preserve traffic flow.
- Case 2: Cut-in. If an adjacent vehicle demonstrates pronounced lateral movement toward the ego lane, the ego vehicle initiates an earlier and smoother response, such as gentle deceleration, to reduce abrupt maneuvers and improve comfort while maintaining safety.
- Case 3: Early Lane Change Away. If the front vehicle moves in the opposite direction of the ego lane, the ego vehicle avoids overly conservative actions to prevent unnecessary traffic disturbance.
- Case 4: Lane Keeping. If the front vehicle remains stably in the ego lane without significant lateral displacement, the ego vehicle follows regular car-following behavior.

By incorporating this predictive decision policy into the overall architecture, the framework establishes an interpretable link between learned trajectory predictions and ego response behavior, enabling a qualitative assessment of comfort and traffic-flow impacts.

EXPERIMENTS AND RESULTS

EXPERIMENTAL DESIGN

In this section, we describe the experimental design used to validate the performance of the key components in the pro-posed framework. The design includes data collection setup, comparison baselines, evaluation metrics, and hyperparameter configurations employed during training and inference. In addition to experiments using real-world dashcam footage, we also conduct simulation-based evaluations to assess the practical impact of the proposed method.

Data Source

The dataset used in this study was col- lected from a motor vehicle only expressway located near Cheonan–Asan, South Korea (about 7 km). Forward-facing dashcam videos were recorded from test vehicles under real- world traffic conditions, capturing diverse interaction scenarios such as merges, cut-ins, and early lane changes. The finalized dataset comprises a total of 261 video clips. To ensure the robustness and diversity of the trajectory prediction under various driving maneuvers, these clips were categorized into 194 lane-keeping (LK) and 67 lane-changing (LC) scenarios. Each clip spans approximately 10 seconds at 30 frames per second (fps), yielding 300 frames of trajectory data per sequence. For model training and evaluation, we implemented an 80/20 split, resulting in a training set of 209 clips and a test set of 52 clips. All object trajectories were extracted through a unified detection and tracking pipeline and synchronized in a top-view coordinate space. The training samples were generated using a sliding window approach, utilizing 30 consecutive frames (1.0 second) as the observation period to predict the subsequent 150 frames.

Baselines

To validate the performance of the proposed TGCA-based trajectory prediction model, we conducted a comprehensive comparison against a broad set of representative baselines categorized into two methodological families: graph-based models and time-series models.

Within the graph-based family, we evaluated the Graph Convolutional Network (GCN) (18), which applies spatial graph convolution operations to capture localized vehicle interactions, and the Temporal Graph Convolutional Network (T-GCN) (19), which integrates GCNs with gated recurrent units to simultaneously model spatial topologies and temporal dynamics. Additionally, we included the Graph Attention Network (GAT) (20), which utilizes attention coefficients for adaptive neighbor weighting; the Temporal Graph Attention Network (TGAT), which extends this mechanism to capture time-varying social interactions; and MixHop (21), which aggregates features across multiple hop distances to encompass both local and long-range spatial relationships.

The time-series family comprises standard sequence modeling approaches, including Long-Short Term Memory (LSTM) (22) for learning temporal dynamics, and Attention LSTM (AT-LSTM) (23), which enhances the standard architecture with temporal attention to emphasize influential historical steps. We also compared against Attention GRU (AT-GRU) (24), combining Gated Recurrent Units with temporal attention for efficient sequence modeling, and the Transformer (25), which employs multi-head self-attention mechanisms to capture long-range dependencies without relying on recurrence.

Evaluation Metrics

To comprehensively assess multi-agent trajectory prediction accuracy, we employ Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) to measure average absolute errors and highlight larger deviations, respectively. To evaluate performance under uncertainty, we utilize Minimum Average Displacement Error (minADE), which calculates the lowest average L2 distance among multiple potential trajectories to reflect the best achievable accuracy across the duration. Similarly, Minimum Final Displacement Error (minFDE) identifies the trajectory with the smallest L2 distance at the final prediction step to quantify endpoint accuracy. Collectively, these metrics capture both average and worst-case prediction quality in deterministic and uncertainty-aware settings.

Implementation Details

All models were trained for 100 epochs using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 64. The input observation window consisted of 30 frames (equivalent to 1 second at 30 fps) processed with a sliding window stride of 1 frame, while the prediction horizon was set to 150 frames (5 seconds). Performance results are reported across prediction horizons $P \in \{1,2,3,4,5\}$ seconds, corresponding to $\{30,60,90,120,150\}$ future frames. For the proposed TGCA model configuration, the hidden dimension d_h was set to 128, and the number of attention heads was fixed at 4.

PERFORMANCE EVALUATION

To validate the effectiveness of the proposed TGCA model, we conducted a comprehensive comparison against a broad range of graph-based and time-series baselines. Performance was

evaluated across prediction horizons $P \in \{1,2,3,4,5\}$ seconds using standard metrics (MAE, RMSE, minADE, and minFDE) on trajectories projected onto a top-view plane and standardized via per-axis z-score normalization (see Table 1).

The results clearly demonstrate that TGCA outperforms all baselines by a large margin. Traditional sequence models (e.g., LSTM, GRU) and even Transformer models showed limited adaptability to localized vehicle interactions, resulting in consistently higher error ranges. In contrast, graph-based models (GCN, GAT, and MixHop) captured vehicle interactions more effectively, with GAT showing notable improvements due to dynamic node weighting; however, even GAT's best-case performance remained significantly inferior to the proposed method. Ultimately, TGCA achieved the best results across all metrics and horizons, recording the lowest MAE and RMSE values of approximately 0.08 and 0.25, respectively. This consistent performance is attributed to TGCA's ability to incorporate temporally directed attention and context-aware edge weighting, enabling a nuanced understanding of both spatial and temporal vehicle interactions.

Base line	*P*	**MAE**	**RMSE**	**Min ADE**	**Min FDE**	**Base line**	*P*	**MAE**	**RMSE**	**Min ADE**	**Min FDE**
AT-LSTM	1	0.43	0.62	2.08	3.26	Transformer	1	0.70	0.99	3.86	4.57
	2	0.76	0.96	3.81	3.89		2	0.85	1.14	4.74	4.64
	3	0.73	0.94	3.70	3.32		3	0.85	1.15	4.77	4.64
	4	0.65	0.87	3.14	2.99		4	0.84	1.14	4.72	4.57
	5	0.70	0.97	3.25	3.52		5	0.84	1.15	4.72	4.68
MixHop	1	0.62	0.92	3.50	4.03	T-GCN	1	0.70	0.99	3.86	4.57
	2	0.73	1.04	4.22	4.06		2	0.85	1.14	4.74	4.64
	3	0.74	1.04	4.24	4.13		3	0.85	1.15	4.77	4.64
	4	0.74	1.05	4.25	4.10		4	0.84	1.14	4.72	4.57
	5	0.74	1.04	4.25	4.22		5	0.84	1.15	4.72	4.68
GAT	1	0.64	0.92	3.57	4.06	TGAT	1	0.89	1.15	4.67	4.48
	2	0.74	1.02	4.22	4.13		2	0.82	1.09	4.56	4.39
	3	0.75	1.03	4.27	4.18		3	0.81	1.08	4.54	4.45
	4	0.75	1.05	4.32	4.21		4	0.82	1.09	4.56	4.45
	5	0.76	1.06	4.35	4.26		5	0.82	1.10	4.60	4.55
LSTM	1	0.49	0.70	2.46	3.98	AT-GRU	1	0.84	1.17	4.24	7.10
	2	0.90	1.13	4.63	4.70		2	1.65	2.05	8.36	8.35
	3	0.87	1.10	4.42	3.94		3	1.52	1.91	7.69	6.92
	4	0.74	0.97	3.69	3.52		4	1.34	1.73	6.51	6.00
	5	0.78	1.06	3.82	4.14		5	1.47	1.98	6.73	7.98

GCN	1	0.67	0.96	3.69	4.29	**Ours**	1	0.09	0.28	0.59	**0.50**
	2	0.78	1.07	4.39	4.23		2	**0.08**	0.26	**0.57**	**0.50**
	3	0.77	1.05	4.35	4.24		3	**0.08**	**0.25**	**0.57**	0.51
	4	0.77	1.05	4.35	4.26		4	**0.08**	0.27	0.58	0.51
	5	0.78	1.07	4.40	4.33		5	0.09	0.29	0.63	0.53

Table 1. Results of the proposed model compared with baselines

DISCUSSION

This study presents a front vehicle trajectory prediction framework employing TGCA to model spatio-temporal interactions, enabling context-aware and proactive driving strategies by selectively attending to influential vehicles. To validate the framework, data-driven trajectory prediction experiments were conducted using monocular dashcam footage collected in real-world conditions, ensuring algorithmic soundness.

Naturally, the monocular vision-based architecture carries inherent limitations, such as restricted field of view and susceptibility to occlusion, which may reduce reliability for long-horizon forecasting compared to multi-sensor or map-based predictors. Additionally, the study improves comfort indirectly through prediction-driven smoothing rather than explicit objective optimization. Future iterations could integrate multi-modal sensor fusion (e.g., radar, LiDAR) and HD maps to improve robustness and long-term plausibility. Nevertheless, this work differentiates itself by demonstrating that proactive prediction under realistic sensing conditions yields meaningful benefits.

Regarding real-time deployment, runtime analysis on an NVIDIA RTX 3070 environment yielded an average end-to-end inference speed of 7.19 fps. Decomposition reveals that the trajectory prediction module (approximately 10.3 fps) is not the dominant bottleneck compared to the perception front-end. Since the TGCA module computes cross-attention within a short sliding window, prediction costs remain moderate, suggesting that real-time operation is achievable with system-level optimization. Ultimately, this study contributes meaningful insights into vision-only, lead-centric interaction modeling that explicitly targets earlier and smoother ego responses.

CONCLUSION

This study proposed a predictive driving framework to support anticipatory maneuver planning in advanced AD systems. By leveraging monocular dashcam footage, the framework forecasts the trajectories of front vehicles and dynamically models spatio-temporal interactions through the introduction of a Temporal Graph Cross Attention (TGCA) mechanism. Designed to selectively attend to vehicles that exert substantial influence on the ego vehicle's behavior, TGCA enables context-aware and proactive driving decisions even under complex traffic conditions.

TGCA was introduced to address the limitations of conventional self-attention and single-frame-based graph models. By adopting a cross-attention structure across consecutive frames, the model facilitates temporal information flow, allowing early detection of behavioral changes among surrounding vehicles. Furthermore, the integration of edge weights modulates attention scores based on spatial proximity and contextual relevance, effectively filtering out irrelevant agents. Specifically, the full TGCA model achieved a 5-second prediction MAE of approximately 0.09, RMSE of 0.29, and minFDE of 0.63, reflecting over 85.5% error reduction compared to GAT.

In conclusion, this study establishes a new anticipatory driving paradigm by formalizing and validating the problem of front vehicle trajectory prediction, which has been largely overlooked in prior research. Despite the inherent constraints of monocular vision—such as limited field of view and occlusion—the proposed approach achieves reliable predictive performance and practical driving improvements. These findings suggest that the framework may serve as a foundational reference for cost-effective and scalable integration of prediction-perception modules in future autonomous driving systems.

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